MLOPS ASSIGNMENT 2

Sentiment Inference Experiment using MLflow

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Task 4: Short Report

1. Which run had the lowest latency, and why might that be?

The **run_batch4** run had the lowest latency at **0.157 seconds**. It was faster than run_batch8 (0.184 seconds) and run_batch1 (0.341 seconds). A batch size of 4 may have given a good balance between processing several sentences together and the overhead of each inference call. The result is based on this small 10-sentence dataset and may differ with a larger dataset.

2. Did accuracy change between runs? Why or why not?

No, accuracy remained **1.00 (100%)** in all three runs. Each run used the same DistilBERT sentiment model and the same 10 test sentences. Changing the batch size changes how the sentences are grouped for processing, but it does not normally change the model weights or the expected predictions. Therefore, the measured accuracy stayed the same.

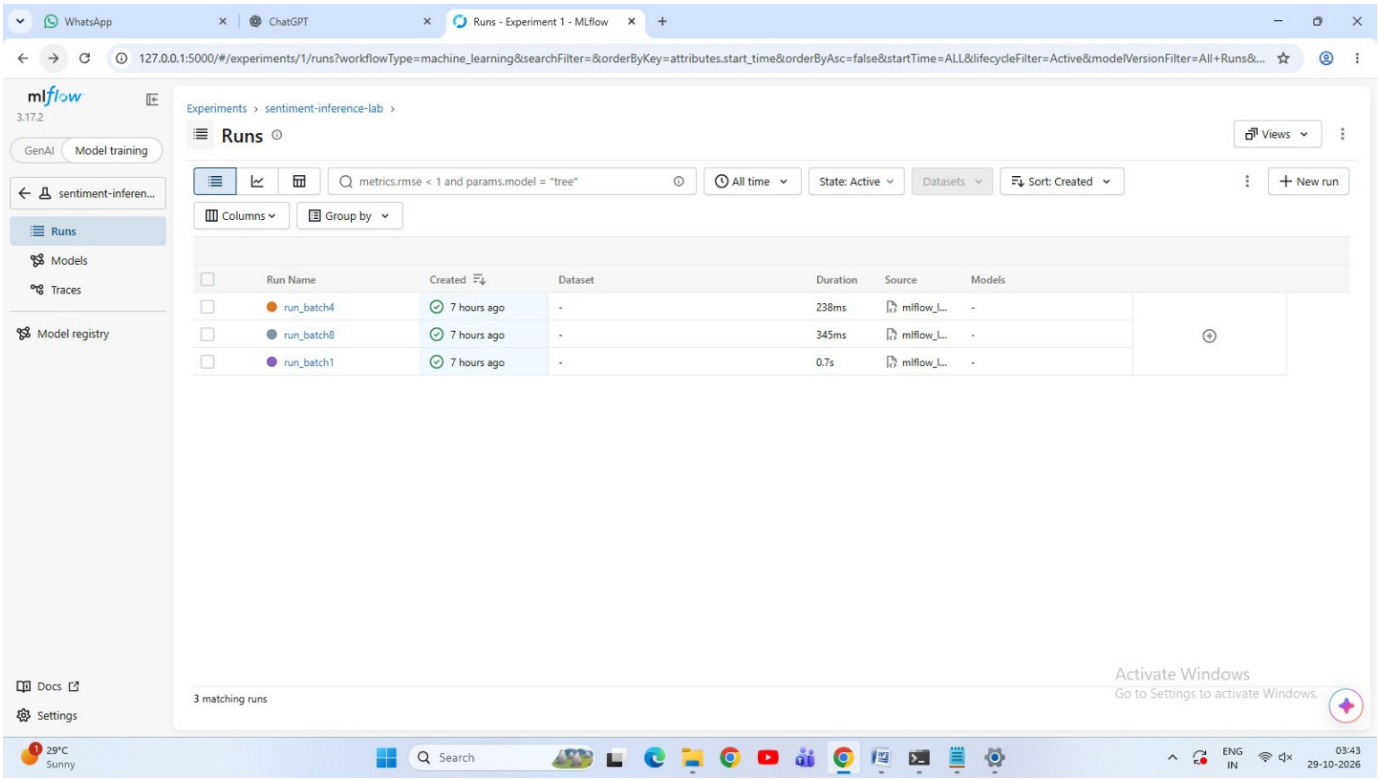
3. Name two benefits of using MLflow here instead of printing results in the terminal.

First, MLflow saves each run's parameters, metrics and output artifacts, so the results can be checked later instead of being lost in terminal output. Second, MLflow makes it easy to compare different runs in one place using tables and charts. This helps us see how changes such as batch size affect latency and other metrics. It also makes experiment tracking more organised.

Recorded results: run_batch1 — batch size 1, latency 0.341 s; run_batch8 — batch size 8, latency 0.184 s; run_batch4 — batch size 4, latency 0.157 s. Accuracy was 1.00 for all runs.

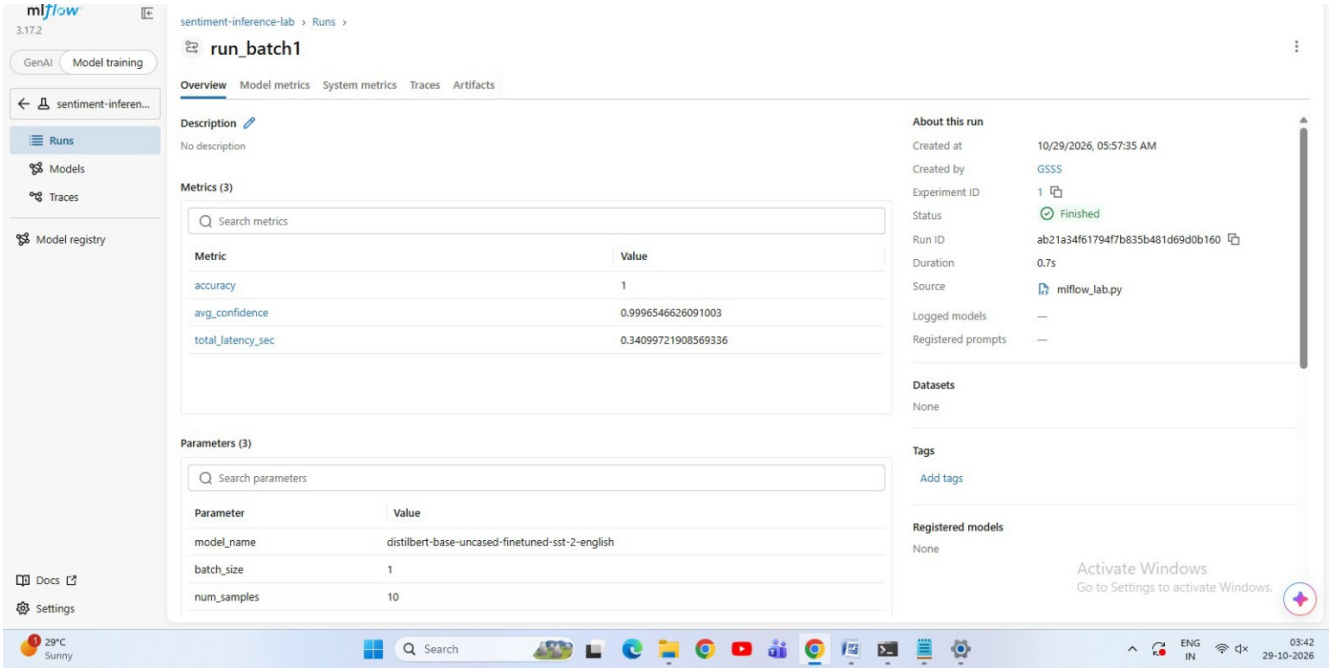
Screenshot of the MLflow runs table

The experiment contains three completed runs: run_batch1, run_batch8 and run_batch4.

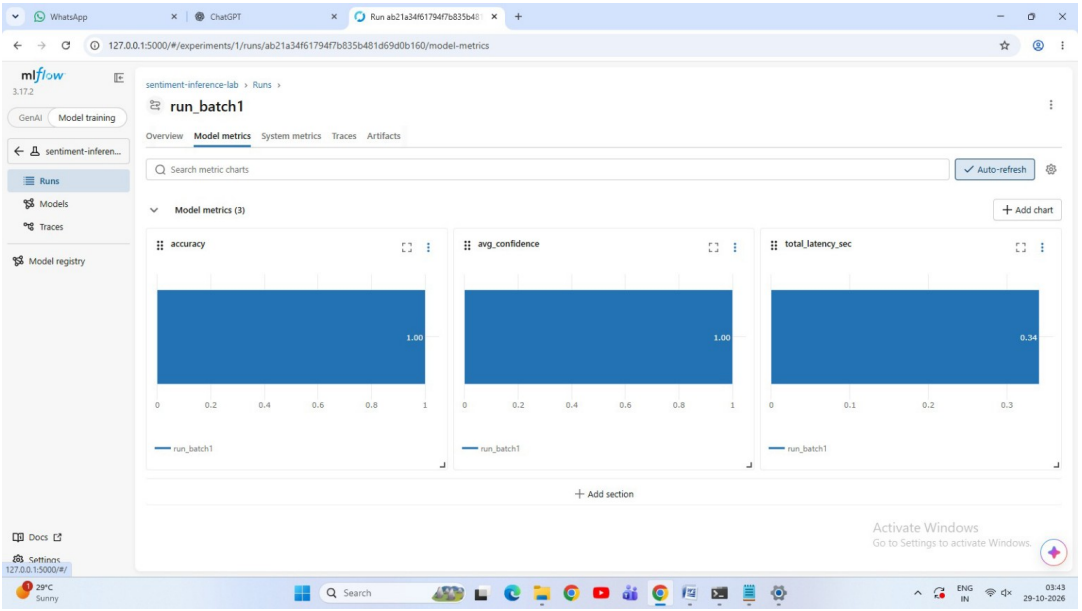


Run details: run_batch1

Batch size 1 — accuracy 1.00 and total latency approximately 0.341 seconds.

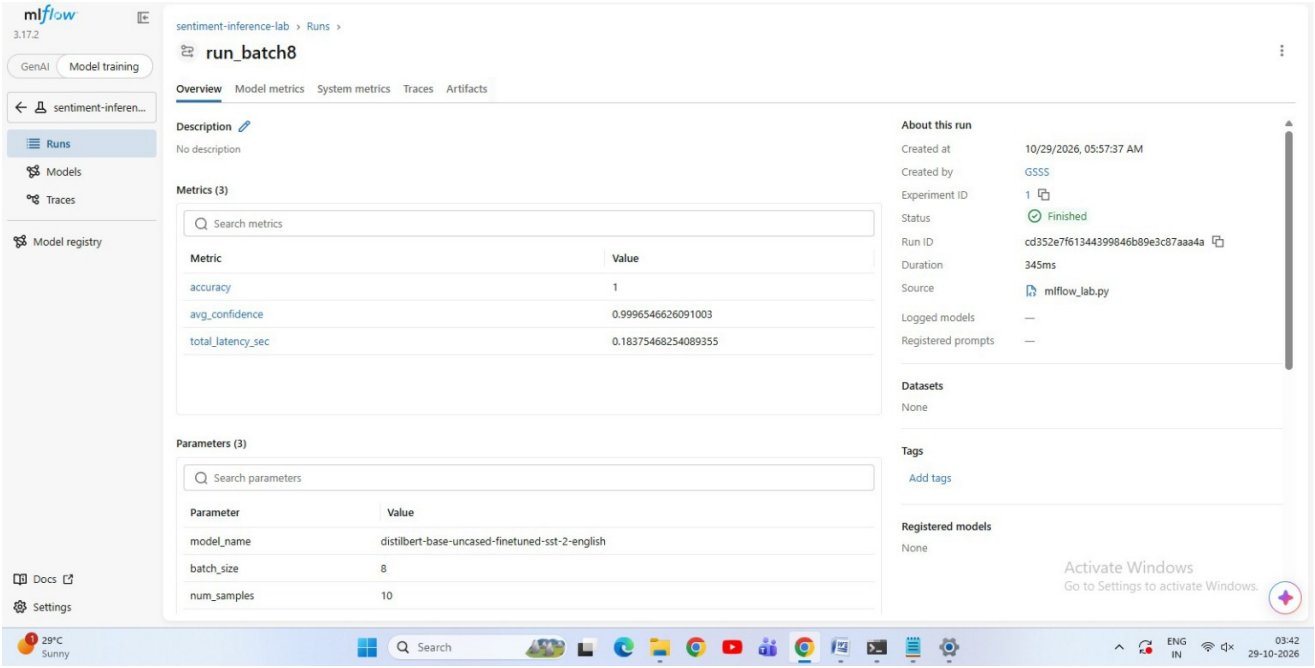


Model metrics

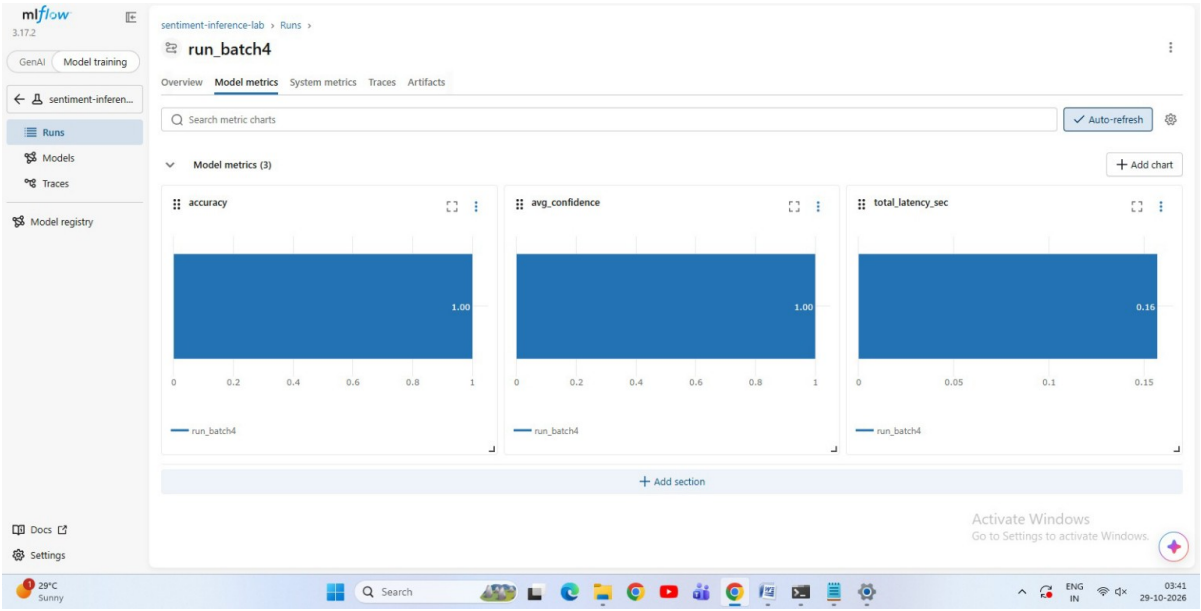


Run details: run_batch8

Batch size 8 — accuracy 1.00 and total latency approximately 0.184 seconds.

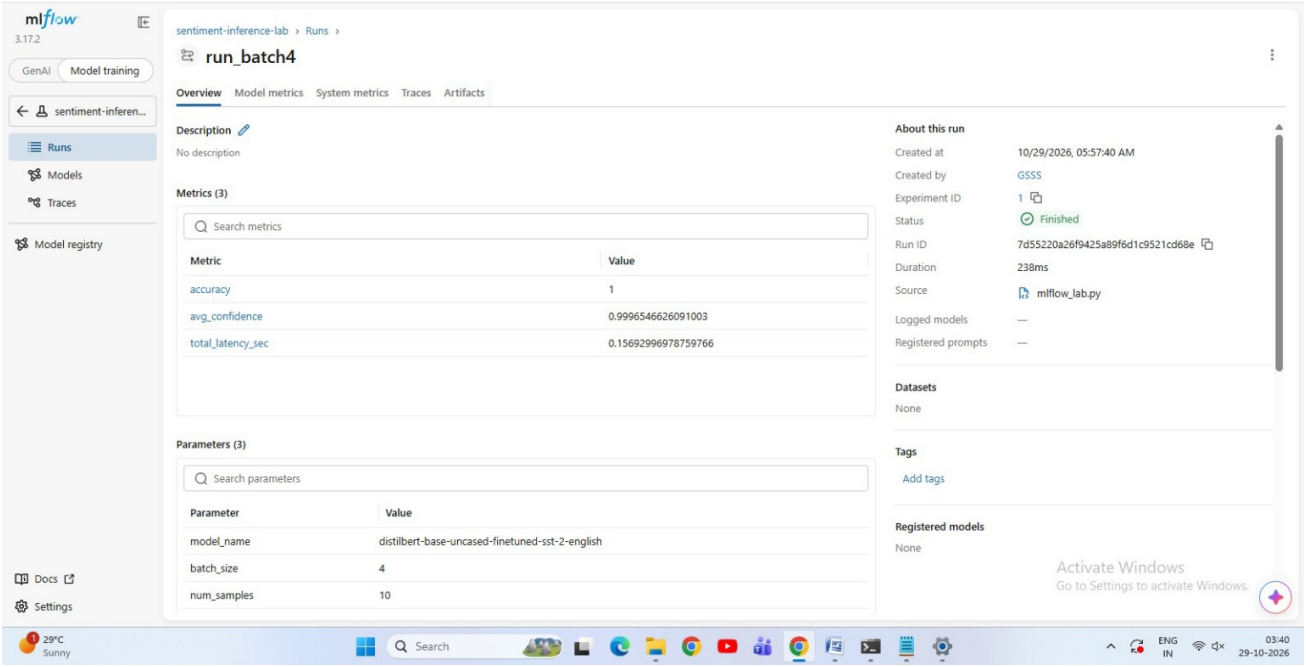


Model metrics

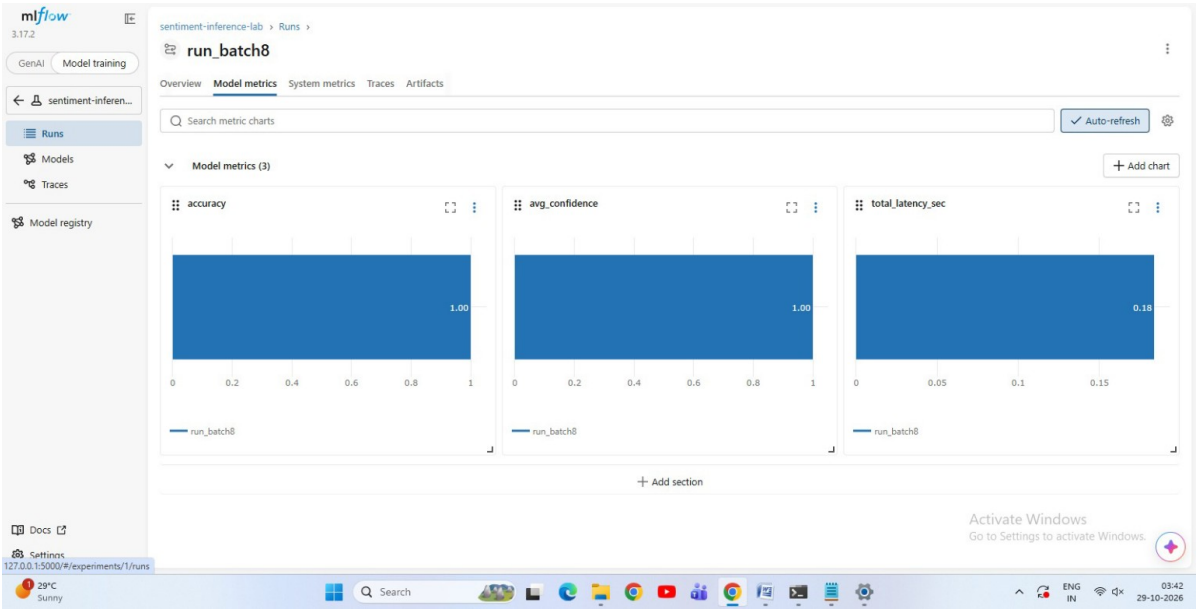


Run details: run_batch4

Batch size 4 — accuracy 1.00 and total latency approximately 0.157 seconds.

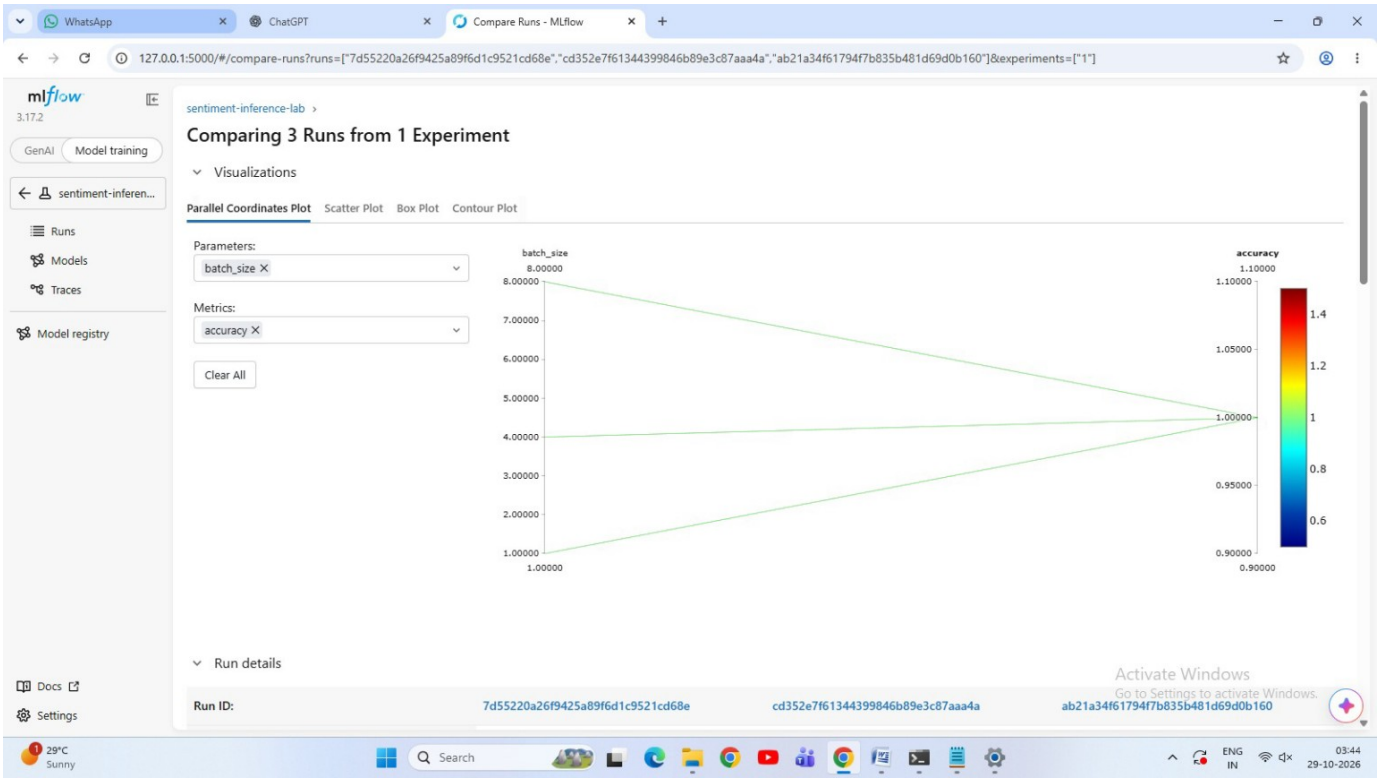


Model metrics



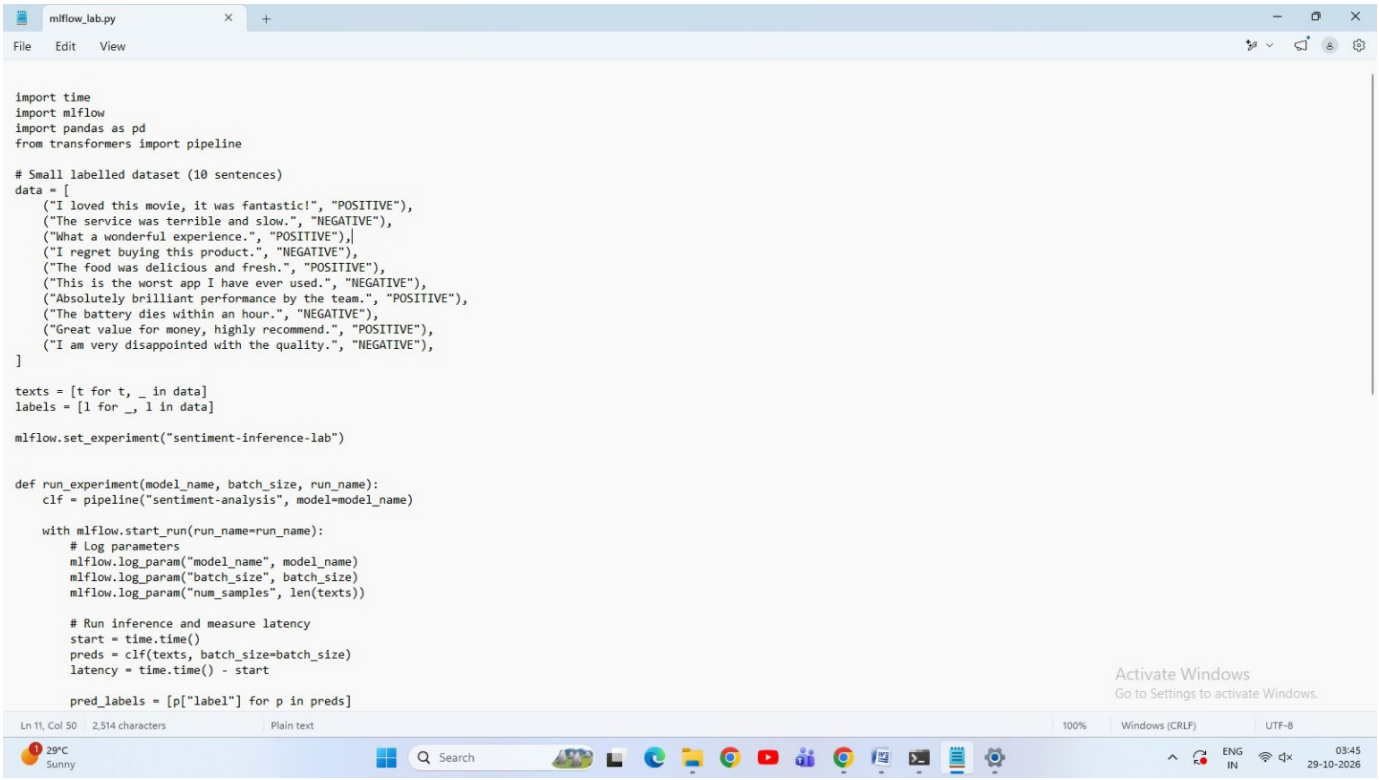
Screenshot of the Compare view

The comparison view displays the three runs in the same experiment.



mlflow_lab.py file

Screenshot of the Python script used to run the sentiment inference experiments and log results to MLflow.



```
import time
import mlflow
import pandas as pd
from transformers import pipeline

# Small labelled dataset (10 sentences)
data = [
    ("I loved this movie, it was fantastic!", "POSITIVE"),
    ("The service was terrible and slow.", "NEGATIVE"),
    ("What a wonderful experience.", "POSITIVE"),
    ("I regret buying this product.", "NEGATIVE"),
    ("The food was delicious and fresh.", "POSITIVE"),
    ("This is the worst app I have ever used.", "NEGATIVE"),
    ("Absolutely brilliant performance by the team.", "POSITIVE"),
    ("The battery dies within an hour.", "NEGATIVE"),
    ("Great value for money, highly recommend.", "POSITIVE"),
    ("I am very disappointed with the quality.", "NEGATIVE"),
]

texts = [t for t, _ in data]
labels = [l for _, l in data]

mlflow.set_experiment("sentiment-inference-lab")


def run_experiment(model_name, batch_size, run_name):
    clf = pipeline("sentiment-analysis", model=model_name)

    with mlflow.start_run(run_name=run_name):
        # Log parameters
        mlflow.log_param("model_name", model_name)
        mlflow.log_param("batch_size", batch_size)
        mlflow.log_param("num_samples", len(texts))

        # Run inference and measure latency
        start = time.time()
        preds = clf(texts, batch_size=batch_size)
        latency = time.time() - start

        pred_labels = [p["label"] for p in preds]
```

Execution

The terminal output shows the three runs completing and MLflow starting successfully.

